- After incubation, plates are observed for growth of colonies (based on the streaking pattern and density of culture growth of microbes are abundant in the first sector in comparison with the formation of separated discrete colonies in the fourth sector of the agar medium).
- Each isolated colony is assumed to be grown from a single bacteria and thus represent a clone of pure culture.
- Successful isolation depends on spatial separation of single cells (Figure 5.12).

Infobits

Micro manipulator: It is a device used along with a microscope to pick a single bacterial cell from a mixed culture.

It has micropipette or microprobe so that a single cell can be picked up.

5.4 Growth and Colony Characteristics of Bacteria and Fungi

In the previous section we have learned the various types of media and specific purpose of each medium. Morphology is the basic criteria for the isolation, identification and classification of microorganisms. Colony characteristics are the basic tool in the field of taxonomy.

Bacteria grow in both solid and liquid medium, but identification will be easy on the solid medium. In solid medium bacteria form colonies. In liquid medium growth of bacteria are generally not distinctive because there is uniform turbidity or sediment at the bottom or pellicle is formed on the surface.

Some basic attributes such as shape, size, colour, pigmentation, texture, elevation and margin of the bacterial colony in the growth medium are explained below.

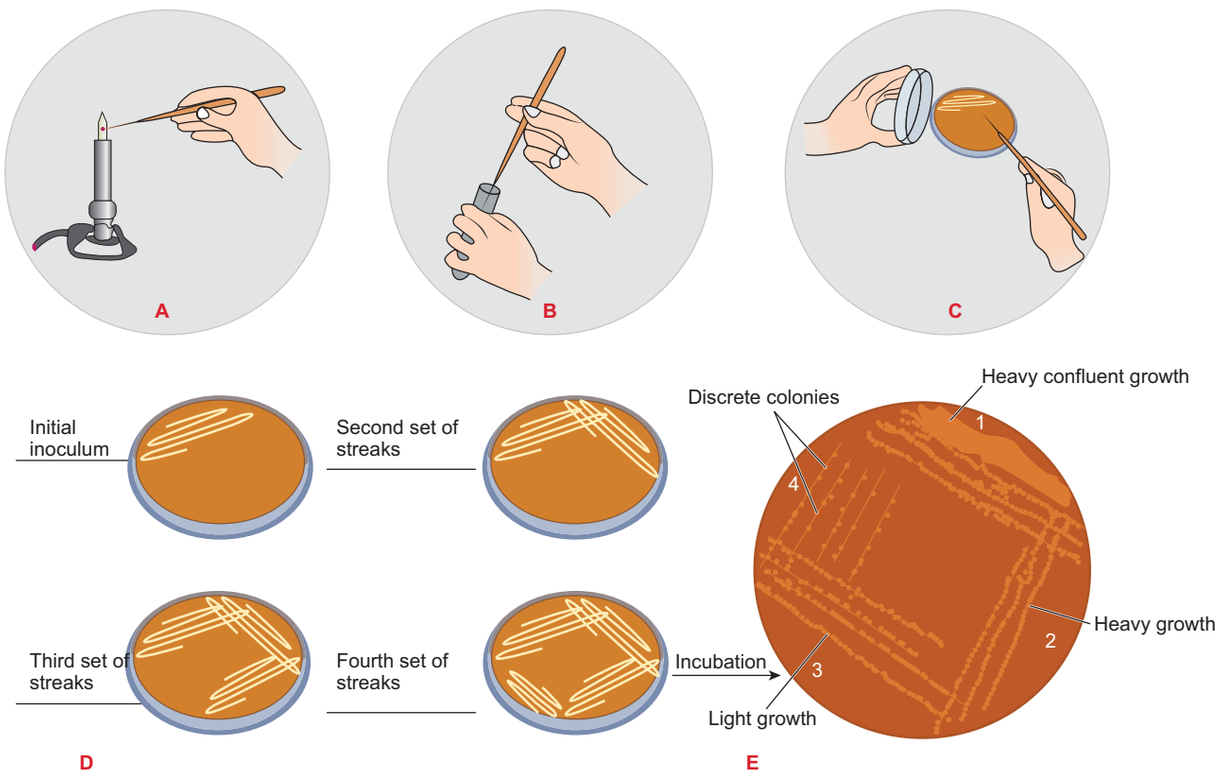


Figure 5.12: Steps in streak plate isolation method

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